CS-GY 6033

Design and Analysis of Algorithms I

Lecture 11 | Part 1

Dynamic Programming

Dynamic Programming

- We’ve seen that dynamic programming can lead to fast algorithms that find the optimal answer.
- Today, we’ll see one data science application: longest common substring.
- Used to match DNA sequences, fuzzy string comparison, etc.

The Strategy

1. Backtracking solution.

2. A “nice” backtracking solution with overlapping subproblems.

3. Memoization.

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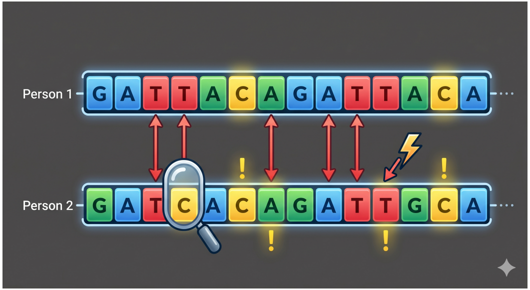
Design and Analysis of Algorithms I

Lecture 11 | Part 2

Longest Common Subsequence

DNA String Matching

- Suppose you're analyzing a genome.
- DNA is a sequence of `G,A,T,C`.
- Mutations cause same gene to have slight differences.
- Person 1: `GATTACAGATTACA`
- Person 2: `GATCACAGTTGCA`



```
lectures/12-dp-lcs/code on  main [!?] via 🐍 v3.10.12 via ❄️
❯ git cmmti
git: 'cmmti' is not a git command. See 'git --help'.

The most similar command is
        commit
```

Measuring Differences

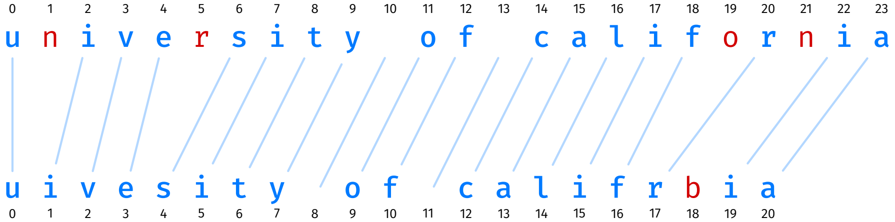
- Given two strings of (possibly) different lengths.
- Measure how similar they are.
- One approach: **longest common subsequences**.

Common Subsequences

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
u	n	i	v	e	r	s	i	t	y		o	f		c	a	l	i	f	o	r	n	i	a

u	i	v	e	s	i	t	y		o	f		c	a	l	i	f	r	b	i	a
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Common Subsequences



Longest Common Subsequences

- We will measure similarity by finding length of the **longest common subsequence** (LCS).

- Now: let’s define the LCS..

Subsequences

```
s a n d i e g o
s a n d i e g o   →   igo
s a n d i e g o   →   sio
s a n d i e g o   →   sadego
s a n d i e g o   →   sandiego
```

Not Subsequences

```
s a n d i e g o
s a n d i e g o    ↛    sea
s a n d i e g o    ↛    sooo
```

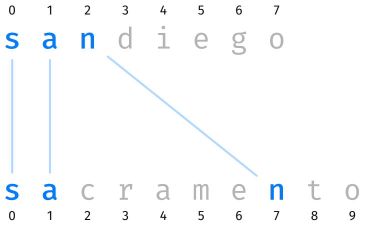
Subsequences

- A **subsequence** of a string `s` of length n is determined by a strictly monotonically increasing sequence of indices with values in $\{0, 1, \ldots, n - 1\}$.

0	1	2	3	4	5	6	7		0	1	3	5	6	7
s	a	n	d	i	e	g	o	→	s	a	d	e	g	o

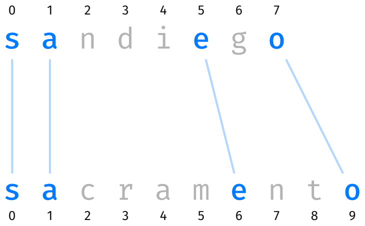
Common Subsequences

- Given two strings, a **common subsequence** is subsequence that appears in both.



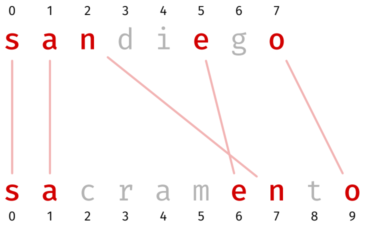
Common Subsequences

- Given two strings, a **common subsequence** is subsequence that appears in both.



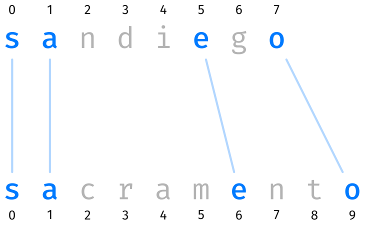
Not Common Subsequences

- The lines cannot overlap.



Longest Common Subsequences

- A **longest common subsequence** (LCS) between two strings is a common subsequence that has the greatest length out of all common subsequences.



Main Idea

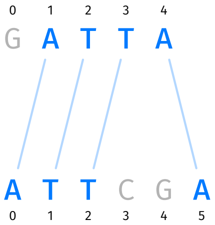
The longer the LCS, the “more similar” the two strings.

Common Subsequences, Formally

- Our backtracking solution will build a common subsequence piece by piece.

- How can we represent the idea of “lines between letters” more formally?

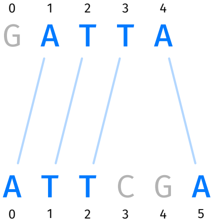
Matching



(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

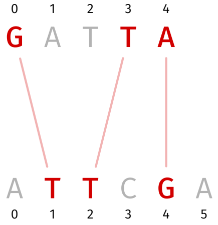
Matching

- A **matching** between strings a and b is a set of (i, j) pairs.
- Each (i, j) pair is interpreted as "`a[i]` is paired with `b[j]`".
- Example: $\{(1, 0), (2, 1), (3, 2), (4, 5)\}$



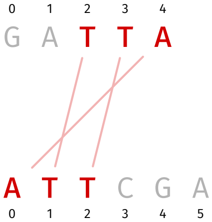
Invalid Matchings

- Not all matchings represent common subsequences!
- Example: $\{(0,1),(3,2),(4,4)\}$:



Invalid Matchings

- Not all matchings represent common subsequences!
- Example: $\{(4,0),(2,1),(3,2)\}$:

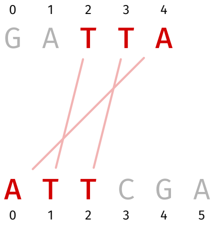


Valid Matchings

- We'll say a matching M is **valid** if:
  - `a[i] == b[j]` for every pair (i, j); and
  - there are no "crossed lines"

“Crossed Lines”

- Suppose (i, j) and (i', j') are in the matching.
- “Crossed lines” occur when either:
  - $i < i'$ but $j \geq j'$; or
  - $i > i'$ but $j \leq j'$.

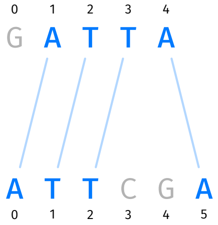


Valid Matchings

- We'll say a matching M is **valid** if:
  - `a[i] == b[j]` for every pair (i, j); and
  - there are no "crossed lines". that is, for every choice of distinct pairs $(i, j), (i', j') \in M$:

$$i < i' \text{ and } j < j' \qquad \text{or} \qquad i > i' \text{ and } j > j'$$

- Example: $\{(1, 0), (2, 1), (3, 2), (4, 5)\}$



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Design and Analysis of Algorithms I

Lecture 11 | Part 3

Step 01: Backtracking

Road to Dynamic Programming

- We’ll follow same road to a DP solution as last time.
- **Step 01: Backtracking solution.**
- Step 02: A “nice” backtracking solution with overlapping subproblems.
- Step 03: Memoization.

Backtracking

- We'll build up a matching, one pair at a time.
- Choose an arbitrary pair, `(i, j)`.
  - Recursively see what happens if we **do** include `(i, j)`.
  - Recursively see what happens if we **don't** include `(i, j)`.
- This will try **all valid matchings**, keep the best.

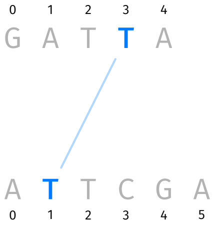
Backtracking

```python
def lcs_bt(a, b, pairs):
    """Solve find best matching using the pairs in `pairs`."""
    pair = pairs.arbitrary_pair()

    if pair is None:
        return 0

    i, j = pair

    # best with
    best_with = ...

    # best without
    best_without = ...

    return max(best_with, best_without)
```

Recursive Subproblems

- What is BEST(`a, b, pairs`) if we assume that `(i, j)` **is** in matching?
- If `a[i] != b[j]`:
  - Your current common substring is **invalid**. Length is zero.
  - Don't build matching further.
- If `a[i] == b[j]`:
  - Your current common substring has length one.
  - Pairs remaining to choose from: those **compatible** with (i, j).
  - You find yourself in a similar situation as before.
  - Answer: 1 + BEST(`pairs.compatible_with(x)`)

`pairs.compatible_with(x)`



(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

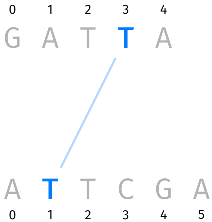
Backtracking

```python
def lcs_bt(a, b, pairs):
    """Solve find best matching using the pairs in `pairs`."""
    pair = pairs.arbitrary_pair()

    if pair is None:
        return 0

    i, j = pair

    # best with
    if a[i] == b[j]:
        best_with = 1 + lcs_bt(a, b, pairs.compatible_with(i, j))
    else:
        best_with = 0

    # best without
    best_without = ...

    return max(best_with, best_without)
```

Recursive Subproblems

- What is BEST(`a, b, pairs`) if we assume that `(i, j)` **is not** in matching?
- Imagine not choosing x.
    - Your current common substring is empty.
    - Pairs left to choose from: all except `(i, j)`.
- You find yourself in a similar situation as before.
- Answer: BEST(a, b, `pairs.without(i, j)`)

`pairs.without(x)`



(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

Backtracking

```python
def lcs_bt(a, b, pairs):
    """Solve find best matching using the pairs in `pairs`."""
    pair = pairs.arbitrary_pair()

    if pair is None:
        return 0

    i, j = pair

    # best with
    # assume (i, j) is in the LCS, but only if a[i] == b[j]
    if a[i] != b[j]:
        best_with = 0
    else:
        best_with = 1 + lcs_bt(a, b, pairs.compatible_with(i, j))

    # best without
    best_without = lcs_bt(a, b, pairs.without(i, j))

    return max(best_with, best_without)
```

Backtracking

- This will try all **valid** matchings.
- Guaranteed to find optimal answer.
- But takes exponential time in worst case.

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Design and Analysis of Algorithms I

Lecture 11 | Part 4

Step 02: A “Nicer” Backtracking Solution

Arbitrary Sets

- In previous backtracking solution, subproblems are arbitrary sets of pairs.
- Rarely see the same subproblem twice.
- This is not good for memoization!

(0,0)	(0,1)	(0,2)	(0,3)	(0,4)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)

Nicer Subproblems

- In backtracking, we are building a solution piece-by-piece.
- In last lecture, we saw that a careful choice of next piece led to nice subproblems.
- Let's try choosing the *last* remaining letters from each string as the next piece of the matching.

Last Letters

0	1	2	3	4	
G	A	T	T	A	

A	T	T	C	G	A
0	1	2	3	4	5

(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

Nicer Backtracking

```python
def lcs_bt_nice(a, b, pairs):
    """Solve find best matching using the pairs in `pairs`."""
    pair = pairs.last_pair()

    if pair is None:
        return 0

    i, j = pair

    # best with
    if a[i] != b[j]:
        best_with = 0
    else:
        best_with = 1 + lcs_bt_nice(a, b, pairs.compatible_with(i, j))

    # best without
    best_without = lcs_bt_nice(a, b, pairs.without(i, j))

    return max(best_with, best_without)
```

Subproblems

- There are two subproblems: LCS using `pairs.compatible_with(i, j)` and LCS using `pairs.without(i, j)`
- Are they “nicer”?

```
pairs.compatible_with(i, j)
```

0	1	2	3	4	
G	A	T	T	A	
A	T	T	C	G	A
0	1	2	3	4	5

(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

Nicer Subproblems

- By taking (i, j) as bottom-right pair, `pairs.compatible_with(i, j)` is again rectangular.
- Easily described by its bottom-right pair, $(i - 1, j - 1)$!
- Instead of keeping set of `pairs`, just need to pass in i and j of last element.

```python
def lcs_bt_nice_2(a, b, i, j):
    """Solve LCS problem for a[:i], b[:j]."""
    if i < 0 or j < 0:
        return 0

    # best with
    if a[i] != b[j]:
        best_with = 0
    else:
        best_with = 1 + lcs_bt_nice_2(a, b, i-1, j-1)

    # best without
    best_without = ...

    return max(best_with, best_without)
```

`pairs.without(i, j)`

0	1	2	3	4	
G	A	T	T	A	
A	T	T	C	G	A
0	1	2	3	4	5

(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

Problem

- `pairs.without(i, j)` is **not** rectangular.
- Cannot be described by a single pair.
- But there's a fix.

Observation

- A common substring cannot have pairs both in the last row and the last column. **Crossing lines!**

0	1	2	3	4
G	A	T	T	A

A	T	T	C	G	A
0	1	2	3	4	5

(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

Consequence

- BEST(`pairs.without(i, j)`) = max {BEST(`pairs.without_row(i)`), BEST(`pairs.without_col(j)`)}

	0	1	2	3	4
	G	A	T	T	A

A	T	T	C	G	A
0	1	2	3	4	5

(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

Observation

- `pairs.without_row(i)` represented by subprob. $(i - 1, j)$
- `pairs.without_col(j)` represented by subprob. $(i, j - 1)$

0	1	2	3	4	
G	A	T	T	A	
A	T	T	C	G	A
0	1	2	3	4	5

(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)
(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,0)	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

“Nice” Backtracking

```
def lcs_bt_nice_2(a, b, i, j):
    """Solve LCS problem for a[:i], b[:j]."""
    if i < 0 or j < 0:
        return 0

    # best with
    if a[i] != b[j]:
        best_with = 0
    else:
        best_with = 1 + lcs_bt_nice_2(a, b, i-1, j-1)

    # best without
    best_without = max(
            lcs_bt_nice_2(a, b, i-1, j),
            lcs_bt_nice_2(a, b, i, j-1)
            )

    return max(best_with, best_without)
```

One More Observation

- This is fine, but we can do a little better.
- If `a[i] == b[j]`, we can assume (i, j) is in matching – don't need to consider otherwise![1]

0	1	2	3	4	
G	A	T	T	A	
A	T	T	C	G	A
0	1	2	3	4	5

[1] This is true if we chose last pair; not true if choice was arbitrary.

“Nicer” Backtracking

```
def lcs_bt_nice_2(a, b, i, j):
    """Solve LCS problem for a[:i], b[:j]."""
    if i < 0 or j < 0:
        return 0

    # best with
    if a[i] == b[j]:
        # best with (i, j)
        return 1 + lcs_bt_nice_2(a, b, i-1, j-1)
    else:
        # best without (i, j)
        return max(
                lcs_bt_nice_2(a, b, i-1, j),
                lcs_bt_nice_2(a, b, i, j-1)
                )
```

Overlapping Subproblems

- Suppose a and b are of length m and n.
- There are mn possible subproblems.
- Backtracking tree has exponentially-many nodes.
- We will see many subproblems over and over again!

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Design and Analysis of Algorithms I

Lecture 11 | Part 5

Step 03: Memoization

Backtracking

- The backtracking solutions are slow.
- `a = 'CATCATCATCATCATGAAAAAAAA'`
- `b = 'GATTACAGATTACAGATTACA'`
- “Nice” backtracking solution: 8 seconds.

Backtracking

- The backtracking solutions are slow.
- `a = 'CATCATCATCATCATGAAAAAAAA'`
- `b = 'GATTACAGATTACAGATTACA'`
- “Nice” backtracking solution: 8 seconds.
- Memoized solution: 100 microseconds.

```python
def lcs_dp(a, b, i=None, j=None, cache=None):
    """Solve LCS problem for a[:i], b[:j]."""
    if i is None:
        i = len(a) - 1

    if j is None:
        j = len(b) - 1

    if cache is None:
        cache = {}

    if i < 0 or j < 0:
        return 0

    if (i,j) in cache:
        return cache[(i, j)]

    # best with
    if a[i] == b[j]:
        # best with (i, j)
        best = 1 + lcs_dp(a, b, i-1, j-1, cache)
    else:
        # best without (i, j)
        best = max(
                lcs_dp(a, b, i-1, j, cache),
                lcs_dp(a, b, i, j-1, cache)
                )

    cache[(i, j)] = best
    return best
```

Top-Down vs. Bottom-Up

- This is the **top-down** dynamic programming solution.
- It takes time $\Theta(mn)$, where m and n are the string lengths.
- To find a bottom-up iterative solution, start with the easiest subproblem.
- What is it?

Bottom-Up Solution

```
# best with
if a[i] == b[j]:
    # best with (i, j)
    best = 1 + lcs_dp(a, b, i-1, j-1, cache)
else:
    # best without (i, j)
    best = max(
           lcs_dp(a, b, i-1, j, cache),
           lcs_dp(a, b, i, j-1, cache)
           )
```

(0,0)	(0,1)	(0,2)
(1,0)	(1,1)	(1,2)
(2,0)	(2,1)	(2,2)
(3,0)	(3,1)	(3,2)

```python
def lcs_dp_bup(a, b):
    """Compute length of LCS, but bottom-up."""
    # initialize cache
    cache = {}
    for i in range(-1, len(a)):
        cache[(i, -1)] = 0
    for j in range(-1, len(b)):
        cache[(-1, j)] = 0

    # fill cache
    for i in range(len(a)):
        for j in range(len(b)):
            if a[i] == b[j]:
                # best with (i, j)
                best = 1 + cache[(i-1, j-1)] # was 1 + lcs_dp(a, b, i-1, j-1, cache)
            else:
                # best without (i, j)
                best = max(
                        cache[(i-1, j)], # was lcs_dp(a, b, i-1, j, cache)
                        cache[(i, j-1)]  # was lcs_dp(a, b, i, j-1, cache)
                        )

            cache[(i, j)] = best

    import pprint
    pprint.pprint(cache)
```

Recoving the Solution

- `lcs_dp` returns the **length** of the LCS.
- How do we recover the actual LCS as a string?
- This information is (implicitly) stored in the cache!

Recovering the Solution

```
a = "ace"
b = "abcde"
```

	-1	0	1	2	3	4
-1	0	0	0	0	0	0
0	0	1	1	1	1	1
1	0	1	1	2	2	2
2	0	1	1	2	2	3

```
# best with
if a[i] == b[j]:
    # best with (i, j)
    best = 1 + lcs_dp(a, b, i-1, j-1, cache)
else:
    # best without (i, j)
    best = max(
            lcs_dp(a, b, i-1, j, cache),
            lcs_dp(a, b, i, j-1, cache)
            )
```

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Design and Analysis of Algorithms I

Lecture 11 | Part 6

String Matching in Practice

In Practice

- The **longest common subsequence** is only one way of measuring similarity between strings.
- In fact, LCS is one specific example of an **edit distance**.

Edit Distance

- An **edit** distance is a measure of similarity between two strings.
- It is the minimum number of **edits** required to transform one string into another.
- LCS: only **insert** and **delete** edits allowed.
- **Levenshtein distance**: insert, delete, and **substitute** edits allowed.

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Design and Analysis of Algorithms I

Lecture 11 | Part 7

Next: String Matching

Today's Problem

- Is a needle in a haystack?
- That is, where does the small string p occur in the large string s?

Strings

- An **alphabet** is a set of possible characters.

$$\Sigma = \{G, A, T, C\}$$

- A **string** is a sequence of characters from the alphabet.

"GATTACATACGAT"

Example: Bitstrings

$\Sigma = \{0, 1\}$

"0110010110"

Example: Text (Latin Alphabet)

$\Sigma = \{\mathtt{a}, \ldots, \mathtt{z}, \mathtt{<space>}\}$

”this is a string”

Comparing Strings

- Suppose `s` and `t` are two strings of equal length, m.
- Checking for equality takes worst-case time $\Theta(m)$ time.

```python
def strings_equal(s, t):
    if len(s) != len(t):
        return False
    for i in range(len(s)):
        if s[i] != t[i]:
            return False
    return True
```

String Matching
(Substring Search)

- **Given**: a string, s, and a pattern string p
- **Determine**: all locations of p in s
- Example:

  `s = "GATTACATACG"`   `p = "TAC"`

Naïve Algorithm

- Idea: “slide” pattern p across s, check for equality at each location.

```
def naive_string_match(s, p):
    match_locations = []
    for i in range(len(s) - len(p) + 1):
        window = s[i:i+len(p)]
        if window == p:
            match_locations.append(i)
    return match_locations
```

Time Complexity

```python
def naive_string_match(s, p):
    match_locations = []
    for i in range(len(s) - len(p) + 1):
        window = s[i:i+len(p)]
        if window == p:
            match_locations.append(i)
    return match_locations
```

Exercise

Exactly how many sliding windows are checked, as a formula involving $|s|$ and $|p|$?

Naïve Algorithm

- Worst case: $\Theta\left((|s| - |p| + 1) \times |p|\right)$ time[2]
- Can we do better?

---

[2] The $+1$ is actually important, since if $|p| = |s|$ this should be $\Theta(p)$

Yes!

- There are numerous ways to do better.
- We'll look at one: **Rabin-Karp**.
- Under some assumptions, takes $\Theta(|s| + |p|)$ expected time.
- Not always the fastest, but **easy** to implement, and generalizes to other problems.

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Design and Analysis of Algorithms I

Lecture 11 | Part 8

Rabin-Karp

Idea

- The naïve algorithm performs (potentially many) comparisons of strings of length $|p|$.
- String comparison is slow: $O(|p|)$ time.
- Integer comparison is fast: $\Theta(1)$ time[3].
- Idea: **hash** strings into integers, compare hashes.

[3] As long as the integers are "not too big"

Recall: Hash Functions

- A **hash function** takes in an object and returns a (small) number.
- **Important**: Given the same object, returns same number.
- It may be possible for two different objects to hash to same number. This is a **collision**.

String Hashing

- A **string hash function** takes a string, returns a number.
- Given same string, returns same number.

```
»> string_hash("testing")
32
»> string_hash("something else")
7
»> string_hash("testing")
32
```

Idea

- Slide a “window” across s, like before.
- But don’t compare window to p directly!
- Instead, compare hash of window to hash of p.
  - If **unequal**, then **no match**.
  - If **equal**, then **possible match**, perform full string comparison to verify.

Example

s="ABBABAABBABA" p="BAA"

x	string_hash(x)
AAA	2
AAB	5
ABA	3
BAA	1
ABB	4
BAB	1
BBA	3
BBB	2

Pseudocode

```
def string_match_with_hashing(s, p):
    match_locations = []
    for i in range(len(s) - len(p) + 1):
        if string_hash(s[i:i+len(p)]) == string_hash(p):
            # make sure this isn't a spurious match due to collision
            if s[i:i+len(p)] == p:
                match_locations.append(i)
    return match_locations
```

Exercise

What is the time complexity of this approach?

Time Complexity of Hashing

- Often, we assume hashing takes constant time with respect to the input size.

- But now, we'll be more careful.

- Hashing a string `p` takes $\Theta(|p|)$ time.

Time Complexity

- Comparing (small) integers takes $\Theta(1)$ time.
- But hashing a string p takes $\Theta(|p|)$.
- In this case, overall:
$$\Omega((|s| + |p| + 1) \cdot |p|)$$
- **No better than naïve!**

Idea: Rolling Hashes

- We have hashed each window from scratch.
- But the strings we are hashing change only a little bit.
  - Example: `s = "ozymandias"`, `p = "mandi"`.
- What if, instead of computing hash from scratch, we could "update" the old hash?

```
»> old_hash = rolling_hash("ozymandias", start=0, stop=5)
»> new_hash = rolling_hash("ozymandias", start=1, stop=6, from=old_hash)
```

Rabin-Karp

- This is the idea behing the **Rabin-Karp** string matching algorithm.
- We'll design a special rolling hash function.
- Instead of computing hash "from scratch", it will "update" old hash in $\Theta(1)$ time.

```python
def rabin_karp(s, p):
    hashed_window = string_hash(s, 0, len(p))
    hashed_pattern = string_hash(p, 0, len(p))
    match_locations = []

    if s[0:len(p)] == p:
        match_locations.append(0)

    for i in range(1, len(s) - len(p) + 1):
        # update the hash
        hashed_window = update_string_hash(s, i, i + len(p), hashed_window)

        if hashed_window == hashed_pattern:
            # make sure this isn't a false match due to collision
            if s[i:i + len(p)] == p:
                match_locations.append(i)

    return match_locations
```

Time Complexity

- $\Theta(|p|)$ time to hash pattern.
- $\Theta(1)$ to update window hash, done $\Theta(|s| - |p| + 1)$ times.
- For each collision, $\Theta(|p|)$ time to check.
- If there are k collisions:

$$\Theta(\underbrace{|p|}_{\text{hash pattern}} + \underbrace{|s| - |p| + 1}_{\text{update window hashes}} + \underbrace{k \cdot |p|}_{\text{check collisions}})$$

$$\Theta(\underbrace{|p|}_{\text{hash pattern}} + \underbrace{|s| - |p| + 1}_{\text{update window hashes}} + \underbrace{k \cdot |p|}_{\text{check collisions}})$$

Exercise

What is the worst case situation for Rabin-Karp?

Worst Case

- In worst case, every position results in a collision.
- That is, there are $\Theta(|s|)$ collisions:

$$\Theta(\underbrace{|p|}_{\text{hash pattern}} + \underbrace{|s| - |p| + 1}_{\text{update windows}} + \underbrace{|s| \cdot |p|}_{\text{check collisions}}) \rightarrow \Theta(|s| \cdot |p|)$$

- Example: `s = "aaaaaaaaa", p = "aaa"`
- This is just as **bad** as naïve!

More Realistic Time Complexity

- Typicall, there are only a few valid matches and a few spurious matches.
- Number of collisions depends on hash function.
- Our hash function will reasonably have $\Theta(|s|/|p|)$ collisions.

$$\Theta(\underbrace{|p|}_{\text{hash pattern}} + \underbrace{|s| - |p| + 1}_{\text{update windows}} + \underbrace{c \cdot |p|}_{\text{check collisions}}) \quad \rightarrow \quad \Theta(|s|)$$

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Design and Analysis of Algorithms I

Lecture 11 | Part 9

Rolling Hashes

The Problem

- We need to hash:
  - `s[0:0 + len(p)]`
  - `s[1:1 + len(p)]`
  - `s[2:2 + len(p)]`
  - ...
- A standard hash function takes $\Theta(|p|)$ time per call.
- But these strings overlap. Maybe we can save work?
- Goal: Design hash function that takes $\Theta(1)$ time to "update" the hash.

Strings as Numbers

- Our hash function should take a string, return a number.
- Should be unlikely that two different strings have same hash.
- Idea: treat each character as a digit in a base-$|\Sigma|$ expansion.

Digression: Decimal Number System

- In the standard decimal (base-10) number system, each digit ranges from 0-9, represents a power of 10.

- Example:

$$\texttt{1532}_{10} = (1 \times 10^3) + (5 \times 10^2) + (3 \times 10^1) + (2 \times 10^0)$$

Digression: Binary Number System

- Computers use binary (base-2). Each digit ranges from 0-1, represents a power of 2.
- Example:

$$\begin{aligned} \texttt{10110}_2 &= (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (1 \times 2^1) + (0 \times 2^0) \\ &= \texttt{22}_{10} \end{aligned}$$

Digression: Base-256

- We can use whatever base is convenient. For instance, base-128, in which each digit ranges from 0-127, represents a power of 128.

$$\texttt{12,97,199}_{128} = (12 \times 128^2) + (97 \times 128^1) + (101 \times 128^0)$$
$$= \texttt{209125}_{10}$$

What does this have to do with strings?

- We can interpret a character in alphabet Σ as a digit value in base $|\Sigma|$.
- For example, suppose $\Sigma = \{a, b\}$.
- Interpret a as 0, b as 1.
- Interpret string "babba" as binary string $\texttt{10110}_2$.
- In decimal: $\texttt{10110}_2 = \texttt{22}_{10}$

Main Idea

We have mapped the string `"babba"` to an integer: 22. In fact, this is the *only* string over Σ that maps to 22. Interpreting a string of `a` and `b` as a binary number hashes the string!

General Strings

- What about general strings, like `"I am a string."`?
- Choose some encoding of characters to numbers.
- Popular (if outdated) encoding: ASCII.
- Maps Latin characters, more, to 0-127. So $|\Sigma| = 128$.

ASCII TABLE

Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	*[NULL]*
1	1	1	1	*[START OF HEADING]*
2	2	10	2	*[START OF TEXT]*
3	3	11	3	*[END OF TEXT]*
4	4	100	4	*[END OF TRANSMISSION]*
5	5	101	5	*[ENQUIRY]*
6	6	110	6	*[ACKNOWLEDGE]*
7	7	111	7	*[BELL]*
8	8	1000	10	*[BACKSPACE]*
9	9	1001	11	*[HORIZONTAL TAB]*
10	A	1010	12	*[LINE FEED]*
11	B	1011	13	*[VERTICAL TAB]*
12	C	1100	14	*[FORM FEED]*
13	D	1101	15	*[CARRIAGE RETURN]*
14	E	1110	16	*[SHIFT OUT]*
15	F	1111	17	*[SHIFT IN]*
16	10	10000	20	*[DATA LINK ESCAPE]*
17	11	10001	21	*[DEVICE CONTROL 1]*
18	12	10010	22	*[DEVICE CONTROL 2]*
19	13	10011	23	*[DEVICE CONTROL 3]*
20	14	10100	24	*[DEVICE CONTROL 4]*
21	15	10101	25	*[NEGATIVE ACKNOWLEDGE]*
22	16	10110	26	*[SYNCHRONOUS IDLE]*
23	17	10111	27	*[ENG OF TRANS. BLOCK]*
24	18	11000	30	*[CANCEL]*
25	19	11001	31	*[END OF MEDIUM]*
26	1A	11010	32	*[SUBSTITUTE]*
27	1B	11011	33	*[ESCAPE]*
28	1C	11100	34	*[FILE SEPARATOR]*
29	1D	11101	35	*[GROUP SEPARATOR]*
30	1E	11110	36	*[RECORD SEPARATOR]*
31	1F	11111	37	*[UNIT SEPARATOR]*
32	20	100000	40	*[SPACE]*
33	21	100001	41	!
34	22	100010	42	"
35	23	100011	43	#
36	24	100100	44	$
37	25	100101	45	%
38	26	100110	46	&
39	27	100111	47	'
40	28	101000	50	(
41	29	101001	51	)
42	2A	101010	52	*
43	2B	101011	53	+
44	2C	101100	54	,
45	2D	101101	55	-
46	2E	101110	56	.
47	2F	101111	57	/

Decimal	Hexadecimal	Binary	Octal	Char
48	30	110000	60	0
49	31	110001	61	1
50	32	110010	62	2
51	33	110011	63	3
52	34	110100	64	4
53	35	110101	65	5
54	36	110110	66	6
55	37	110111	67	7
56	38	111000	70	8
57	39	111001	71	9
58	3A	111010	72	:
59	3B	111011	73	;
60	3C	111100	74	<
61	3D	111101	75	=
62	3E	111110	76	>
63	3F	111111	77	?
64	40	1000000	100	@
65	41	1000001	101	A
66	42	1000010	102	B
67	43	1000011	103	C
68	44	1000100	104	D
69	45	1000101	105	E
70	46	1000110	106	F
71	47	1000111	107	G
72	48	1001000	110	H
73	49	1001001	111	I
74	4A	1001010	112	J
75	4B	1001011	113	K
76	4C	1001100	114	L
77	4D	1001101	115	M
78	4E	1001110	116	N
79	4F	1001111	117	O
80	50	1010000	120	P
81	51	1010001	121	Q
82	52	1010010	122	R
83	53	1010011	123	S
84	54	1010100	124	T
85	55	1010101	125	U
86	56	1010110	126	V
87	57	1010111	127	W
88	58	1011000	130	X
89	59	1011001	131	Y
90	5A	1011010	132	Z
91	5B	1011011	133	[
92	5C	1011100	134	\
93	5D	1011101	135	]
94	5E	1011110	136	^
95	5F	1011111	137	_

Decimal	Hexadecimal	Binary	Octal	Char
96	60	1100000	140	`
97	61	1100001	141	a
98	62	1100010	142	b
99	63	1100011	143	c
100	64	1100100	144	d
101	65	1100101	145	e
102	66	1100110	146	f
103	67	1100111	147	g
104	68	1101000	150	h
105	69	1101001	151	i
106	6A	1101010	152	j
107	6B	1101011	153	k
108	6C	1101100	154	l
109	6D	1101101	155	m
110	6E	1101110	156	n
111	6F	1101111	157	o
112	70	1110000	160	p
113	71	1110001	161	q
114	72	1110010	162	r
115	73	1110011	163	s
116	74	1110100	164	t
117	75	1110101	165	u
118	76	1110110	166	v
119	77	1110111	167	w
120	78	1111000	170	x
121	79	1111001	171	y
122	7A	1111010	172	z
123	7B	1111011	173	{
124	7C	1111100	174	|
125	7D	1111101	175	}
126	7E	1111110	176	~
127	7F	1111111	177	*[DEL]*

In Python

```
»> ord('a')
97
»> ord('Z')
90
»> ord('!')
33
```

ASCII as Base-128

- Each character represents a number in range 0-127.
- A string is a number represented in base-128.
- Example:

$$
\begin{aligned}
&\texttt{Hello}_{128} \\
&= (72 \times 128^4) \\
&\quad + (101 \times 128^3) \\
&\quad + (108 \times 128^2) \\
&\quad + (108 \times 128^1) \\
&\quad + (111 \times 128^0) \\
&= 19540948591_{10}
\end{aligned}
$$

character	ASCII code
H	72
e	101
l	108
o	111

```python
def base_128_hash(s, start, stop):
    """Hash s[start:stop] by interpreting as ASCII base 128"""
    # this is only pseudo-code!
    p = 0
    total = 0
    while stop > start:
        total += ord(s[stop-1]) * 128**p # <--  : O
        p += 1
        stop -= 1
    return total
```

Rolling Hashes

- We can hash a string x by interpreting it as a number in a different base number system.
- But hashing takes time $\Theta(|x|)$.
- With rolling hashes, it will take time $\Theta(1)$ to "update".

character	ASCII code
H	72
e	101
l	108
o	111

Example

- Hash of “Hel” in “**Hel**lo”
- Hash of “ell” in “H**ell**o”

“Updating” a Rolling Hash

- Start with old hash, subtract character to be removed.
- “Shift” by multiplying by 128.
- Add new character.
- Takes $\Theta(1)$ time.

```
def update_base_128_hash(s, start, stop, old):
    # assumes ASCII encoding, base 128
    length = stop - start
    removed_char = ord(s[start - 1]) * 128**(length - 1)
    added_char = ord(s[stop - 1])
    return (old - removed_char) * 128 + added_char
```

```
»> base_128_hash(”Hello”, 0, 3)
1192684
»> base_128_hash(”Hello”, 1, 4)
1668716
»> update_base_128_hash(”Hello”, 1, 4, 1192684)
1668716
```

Note

- In this hashing strategy, there are no collisions!
- Two different string have two different hashes.
- But as we'll see... it isn't practical.

Rabin-Karp

```
def rabin_karp(s, p):
    hashed_window = base_128_hash(s, 0, len(p), q)
    hashed_pattern = base_128_hash(p, 0, len(p), q)
    match_locations = []

    if s[0:len(p)] == p:
        match_locations.append(0)

    for i in range(1, len(s) - len(p) + 1):
        # update the hash
        hashed_window = update_base_128_hash(s, i, i + len(p), hashed_window)

        # hashes are unique; no collisions
        if hashed_window == hashed_pattern:
            match_locations.append(i)

    return match_locations
```

Example

- `s = "this is a test", p = "is"`
- `hashed_pattern = 13555`

i	s[...]	hashed_window
0	"th"	14952
1	"hi"	13417
2	"is"	13555
3	"s "	14752
4	" i"	4201
5	"is"	13555
6	"s "	14752
7	" a"	4193
8	"a "	12448
9	" t"	4212
10	"te"	14949
11	"es"	13043
12	"st"	14836

Large Numbers

- We're hashing because integer comparison takes $\Theta(1)$ time.
- But this is only true if integers are small enough.
- Our integers can get **very large**.

$$128^{|p|-1}$$

Example

```
»> p = ”University of California”
»> base_128_hash(p, 0, len(p))
250986132488946228262668052010265908722774302242017
```

Large Integers

- In some languages, large integers will overflow.
- Python has arbitrary size integers.
- But comparison no longer takes $\Theta(1)$

Solution

- Use **modular arithmetic**.

- Example:
  `(4 + 7) % 3 = 11 % 3 = 2`

- Results in much smaller numbers.

Idea

- Choose a random prime number > $|m|$.
- Do all arithmetic modulo this number.

Fact

$$
\begin{aligned}
&(c_1 \times b^2 + c_2 \times b + c_3) \mod q \\
&\quad = \left(\left[(c_1 \mod q)b + c_2\right] \mod q\right) b + c_3 \mod q
\end{aligned}
$$

- Example: $c_1 = 7, c_2 = 3, c_3 = 5, b = 2, q = 11$
- This allows us to keep numbers small.

```python
import math

def base_128_hash(s, start, stop, q):
    """Hash s[start:stop] mod q by interpreting as ASCII base 128"""
    total = 0
    while stop > start:
        total *= 128
        total += ord(s[start])
        total %= q
        start += 1
    return total
```

```python
def update_base_128_hash(s, start, stop, old, q):
    length = stop - start

    # assumes ASCII encoding, base 128
    # remove the old value, effectively subtracting
    # ord(s[start]) * 128**(length-1) from old, but
    # mod q so that we don't overflow
    new = old - ord(s[start-1]) * pow(128, length - 1, q)

    # "shift" up
    new *= 128
    new %= q

    # add the new character
    new += ord(s[stop - 1])
    new %= q

    return new

print(base_128_hash("DSC190", 0, 6, 499))
```

Example

```
»> base_128_hash("testing", start=0, stop=4, q=117)
103
»> base_128_hash("testing", start=1, stop=5, q=117)
84
»> update_base_128_hash("testing", start=1, stop=5, old=103, q=117)
84
```

Note

- Now there can be collisions!
- Even if window hash matches pattern hash, need to verify that strings are indeed the same.

```python
def rabin_karp(s, p, q):
    hashed_window = base_128_hash(s, 0, len(p), q)
    hashed_pattern = base_128_hash(p, 0, len(p), q)
    match_locations = []

    if s[0:len(p)] == p:
        match_locations.append(0)

    for i in range(1, len(s) - len(p) + 1):
        # update the hash
        hashed_window = update_base_128_hash(s, i, i + len(p), hashed_window, q)

        if hashed_window == hashed_pattern:
            # make sure this isn't a false match due to collision
            if s[i:i + len(p)] == p:
                match_locations.append(i)

    return match_locations
```

Time Complexity

- If q is prime and $> |p|$, the chance of two different strings colliding is small.
- From before: if the number of matches is small, Rabin-Karp will take $\Theta(|s| + |p|)$ expected time.
- Since $|p| \leq |s|$, this is $\Theta(s)$.
- Worst-case time: $\Theta(|s| \cdot |p|)$.